SharedPtr
SharedPtr
vector< smacc2::ISmaccState * >
SmaccStatus
map< std::string, std::shared_ptr< smacc2::ISmaccOrthogonal > >
recursive_mutex
StateMachineInternalAction
map< void *, std::shared_ptr< smacc2::CallbackCounterSemaphore > >
map< std::string, std::pair< std::function< std::string()>, boost::any > >
vector< smacc2_msgs::msg::SmaccTransitionLogEntry >
SMRunMode
int64_t
shared_ptr< smacc2::introspection::SmaccStateInfo >
shared_ptr< smacc2::introspection::SmaccStateMachineInfo >
SharedPtr
#eventsLogPub_
#stateMachinePub_
#stateMachineStatusPub_
#transitionHistoryService_
#transitionLogPub_
#timer_
#currentState_
#status_msg_
#orthogonals_
-eventQueueMutex_
-m_mutex_
-stateMachineCurrentAction
-stateCallbackConnections
-globalData_
-transitionLogHistory_
-runMode_
-stateSeqCounter_
#currentStateInfo_
#stateMachineInfo_
#nh_
smacc2::ISmaccStateMachine
+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ reset()
+ stop()
+ eStop()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfigurationFinished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()
-statusPub_
vector< smacc2::ISmaccUpdatable * >
vector< std::vector< smacc2::ISmaccUpdatable * > >
atomic< int64_t >
double
atomic< bool >
SmaccFifoScheduler *
processor_handle
thread
ExecutionModel
-updatableClients_
-updatableStateElements_
-lastState_
-loop_rate_hz
-end_
-initialized_
-rosInitialized_
-scheduler_
-processorHandle_
-signalDetectorThread_
-executionModel_
-smaccStateMachine_
-signalDetector_
smacc2::SignalDetector
+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ pollingLoop()
+ postEvent()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyRosInitialized()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()
vector< std::shared_ptr< smacc2::ISmaccClient > >
vector< std::vector< std::shared_ptr< smacc2::ISmaccClientBehavior > > >
mutex
int
-stateMachine_
#clients_
-clientBehaviors_
-mutex_
smacc2::ISmaccOrthogonal
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# onInitialize()
# initializeClients()
# assignClientToOrthogonal()
string
std::enable_shared_from_this< SmaccStateInfo >
map< const std::type_info *, std::vector< smacc2::introspection::ClientBehaviorInfoEntry > >
map< const std::type_info *, std::vector< std::shared_ptr< smacc2::introspection::SmaccStateReactorInfo > > >
map< const std::type_info *, std::vector< std::shared_ptr< smacc2::introspection::SmaccEventGeneratorInfo > > >
vector< smacc2::introspection::SmaccTransitionInfo >
vector< std::shared_ptr< smacc2::introspection::SmaccStateInfo > >
type_info *
+parentState_
+stateMachine_
+depth_
+stateIndex_
+demangledStateName
+fullStateName
+staticBehaviorInfo
+stateReactorsInfo
+eventGeneratorsInfo
+transitions_
+children_
+tid_
#node_
smacc2::introspection::SmaccStateInfo
+ SmaccStateInfo()
+ getStateLevel()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()
shared_ptr< rclcpp::Logger >
vector< std::shared_ptr< smacc2::StateReactor > >
vector< std::shared_ptr< smacc2::SmaccEventGenerator > >
-stateMachine_
#logger_
#stateReactors_
#eventGenerators_
#stateInfo_
smacc2::ISmaccState
+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransition()
+ notifyTransitionFromTransitionTypeInfo()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()
#parentState_
-currentState
-stateMachine_
map< smacc2::ComponentKey, std::shared_ptr< smacc2::ISmaccComponent > >
#components_
#stateMachine_
-orthogonal_
smacc2::ISmaccClient
+ ISmaccClient()
+ ~ISmaccClient()
+ onInitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvent()
# onComponentInitialization()
# onStateOrthogonalAllocation()
# createComponent()
# createNamedComponent()
# getNode()
# getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()
boost::signals2::signal< Signature, optional_last_value< typename boost::function_traits< Signature >::result_type >, int, std::less< int >, function< Signature >, typename extended_signature< function_traits< Signature >::arity, Signature >::function_type, boost::signals2::mutex >
boost::signals2::signal< void(int channel, const std::string &error), optional_last_value< typename boost::function_traits< void(int channel, const std::string &error)>::result_type >, int, std::less< int >, function< void(int channel, const std::string &error) >, typename extended_signature< function_traits< void(int channel, const std::string &error)>::arity, void(int channel, const std::string &error) >::function_type, boost::signals2::mutex >
boost::signals2::signal< void(uint8_t states), optional_last_value< typename boost::function_traits< void(uint8_t states) >::result_type >, int, std::less< int >, function< void(uint8_t states)>, typename extended_signature< function_traits< void(uint8_t states)>::arity, void(uint8_t states)>::function_type, boost::signals2::mutex >
smacc2::SmaccSignal< Signature, Combiner, Group, GroupCompare, SlotFunction, ExtendedSlotFunction, Mutex >
boost::signals2::signal< void(int channel, bool state), optional_last_value< typename boost::function_traits< void(int channel, bool state)>::result_type >, int, std::less< int >, function< void(int channel, bool state)>, typename extended_signature< function_traits< void(int channel, bool state)>::arity, void(int channel, bool state)>::function_type, boost::signals2::mutex >
boost::signals2::signal< void(const std::string &), optional_last_value< typename boost::function_traits< void(const std::string &)>::result_type >, int, std::less< int >, function< void(const std::string &)>, typename extended_signature< function_traits< void(const std::string &)>::arity, void(const std::string &)>::function_type, boost::signals2::mutex >
-heartbeat_interval_ms_
-port_
-slave_id_
optional< rclcpp::Duration >
optional< rclcpp::Time >
boost::signals2::signal< void(), optional_last_value< typename boost::function_traits< void()>::result_type >, int, std::less< int >, function< void()>, typename extended_signature< function_traits< void()>::arity, void()>::function_type, boost::signals2::mutex >
-mutex_
-mutex_
-currentOrthogonal
-ip_address_
#owner_
smacc2::ISmaccComponent
+ ISmaccComponent()
+ ~ISmaccComponent()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()
< void(int channel, const std::string &error)>
< void(uint8_t states)>
< void(int channel, bool state)>
< void(const std::string &)>
+COIL_BASE_ADDRESS
+NUM_CHANNELS
< void()>
-periodDuration_
-lastUpdate_
smacc2::SmaccSignal< void(int channel, const std::string &error)>
smacc2::SmaccSignal< void(uint8_t states)>
smacc2::SmaccSignal< void(int channel, bool state)>
smacc2::SmaccSignal< void(const std::string &)>
modbus_t *
function< void()>
smacc2::ISmaccUpdatable
+ ISmaccUpdatable()
+ ISmaccUpdatable()
+ executeUpdate()
+ setUpdatePeriod()
# update()
smacc2::SmaccSignal< void()>
smacc2::ISmaccClientBehavior
+ ISmaccClientBehavior()
+ ~ISmaccClientBehavior()
+ getStateMachine()
+ getName()
+ requiresClient()
+ requiresComponent()
+ requiresComponent()
+ onEntry()
+ onExit()
+ executeOnEntry()
+ executeOnExit()
# runtimeConfigure()
# postEvent()
# postEvent()
# getCurrentState()
# dispose()
# getNode()
# getLogger()
- onStateOrthogonalAllocation()
bool
optional< std::future< int > >
+onConnectionError_
-ctx_
-postConnectionLostEvent
-postConnectionRestoredEvent_
-postFailureEventFn
-postFinishEventFn
-postSuccessEventFn_
+onConnectionLost_
+onConnectionRestored_
-onFailure_
-onFinished_
-onSuccess_
-connect_on_init_
-connected_
-isShutdownRequested_
-onEntryThread_
-onExitThread_
-channel_
cl_modbus_tcp_relay::CpModbusConnection
+ CpModbusConnection()
+ ~CpModbusConnection()
+ onInitialize()
+ onStateOrthogonalAllocation()
+ connect()
+ disconnect()
+ reconnect()
+ isConnected()
+ getContext()
+ getMutex()
+ getIpAddress()
+ getPort()
+ getSlaveId()
+ onConnectionLost()
+ onConnectionRestored()
+ onConnectionError()
# update()
- declareAndLoadParam()
smacc2::SmaccAsyncClientBehavior
+ onStateOrthogonalAllocation()
+ ~SmaccAsyncClientBehavior()
+ onSuccess()
+ onFinished()
+ onFailure()
+ requestForceFinish()
+ executeOnEntry()
+ executeOnExit()
+ waitOnEntryThread()
+ onSuccess()
+ onFinished()
+ onFailure()
# postSuccessEvent()
# postFailureEvent()
# dispose()
# isShutdownRequested()
- waitFutureIfNotFinished()
+onWriteFailure_
+onReadComplete_
+onWriteSuccess_
-postWriteFailureEvent
-postWriteSuccessEvent_
-connectionComponent_
-connectionComponent_
cl_modbus_tcp_relay::CpModbusRelay
+ CpModbusRelay()
+ ~CpModbusRelay()
+ onInitialize()
+ onStateOrthogonalAllocation()
+ writeCoil()
+ readCoil()
+ writeAllCoils()
+ writeAllCoils()
+ readAllCoils()
+ onWriteSuccess()
+ onWriteFailure()
+ onReadComplete()
- channelToAddress()
- isValidChannel()
-relayComponent_
-relayComponent_
cl_modbus_tcp_relay::ClModbusTcpRelay
+ ClModbusTcpRelay()
+ ~ClModbusTcpRelay()
+ onInitialize()
+ onStateOrthogonalAllocation()
+ getConnectionComponent()
+ getRelayComponent()
-client_
cl_modbus_tcp_relay::CbRelayOff
+ CbRelayOff()
+ ~CbRelayOff()
+ onStateOrthogonalAllocation()
+ onEntry()